



Engineer to Excel



CO5- ASSESSMENT TOOL 2

Technical Presentation

CSA1709 - Artificial Intelligence

GUIDED BY :

Dr.P.Rajaram

PRESENTED BY :

V.Sireesha(192472208)



Engineer to Excel



Questions

1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.



Engineer to Excel

Rubrics for Calculation



Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



4.DATA FLOW & PROCESSING PIPELINE



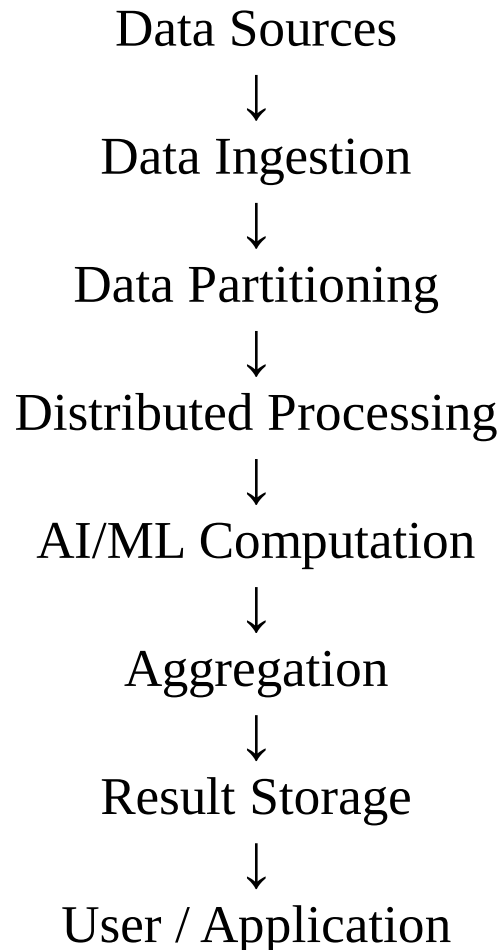
Introduction & Problem

Data Flow and Processing Pipeline in a Distributed AI System

- Distributed AI systems process large datasets using multiple computing resources.
- Data is divided and processed across different components.
- Results are transferred and combined.
- Large datasets can make centralized processing slow.
- Communication can introduce delays.
- Fault tolerance and data consistency are important.

Distributed Processing Pipeline

Processing Pipeline



Explanation

- **Data Sources:** Provide the input data.
- **Data Ingestion:** Collects data into the system.
- **Data Partitioning:** Divides data into smaller parts.
- **Distributed Processing:** Processes different partitions using multiple resources.
- **AI/ML Computation:** Performs intelligent processing.
- **Aggregation:** Combines intermediate results.
- **Result Storage:** Stores the final results.
- **User/Application:** Receives the processed output.



Important Factors



Important Factors:

- Data partitioning
 - Parallel processing
 - Network communication
 - Fault tolerance
 - Resource utilization
 - Result aggregation
- **Data Partitioning** – Divides large datasets for parallel processing.
 - **Parallel Processing** – Processes multiple data parts simultaneously.
 - **Network Communication** – Transfers data between processing components.
 - **Fault Tolerance** – Handles failures during processing.
 - **Resource Utilization** – Efficiently uses available computing resources.
 - **Result Aggregation** – Combines processed results.



Tools



- **PySpark** – Used to process large data.
- **RDD** – Used to store and process data.
- **Spark Monitoring** – Used to check system performance.
- **Logs** – Used to find errors and activities.
- **Performance Monitoring** – Used to check processing speed.
- **Data-Flow Diagram** – Shows how data moves in the system.



Conclusion



- Distributed processing handles large amounts of data efficiently.
- Data is divided and processed in parallel.
- PySpark and RDD help improve processing speed.
- Proper data flow improves system performance and scalability.

Thank

You....